Fiscal Year:	FY 2006	**Task Last Updated:**	FY 07/26/2006
PI Name:	Blakely, Eleanor A Ph.D.		
Project Title:	Early Markers of Space Radiation Induced Cataractogenesis		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	94720	**Congressional District:**	13
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	02-OBPR-02
Start Date:	06/01/2003	**End Date:**	09/30/2007
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received no-cost extension to 9/30/2007 per P. Denkins/JSC (5/07)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	T-465-X		
Performance Goal No.:			
Performance Goal Text:			
	Cataract incidence in astronauts has been correlated with exposure to relatively low absorbed doses of radiation in space flight. The type of space radiation linked with low-dose cataractogenesis has not yet been identified. The goal of this proposal is to test the hypothesis that radiation-induced alterations in early molecular signaling in cultured human lens epithelial (HLE) cells can be used to identify early markers of relative risk of cataracts from key high-LET radiation components. We have shown that exposure to low fluences of high-LET radiation promotes severe lens cell dysfunction via events that include modulation of FGF-2, inhibition of cell cycle control via cyclin-dependent-kinase inhibitors (CDKIs) altered expression of cell receptor and adhesion molecules, and altered apoptotic/enucleation program machinery, in the HLE and human lens fiber (HLF) cells. These observations are consistent with a model in which ionizing radiation initiates premature differentiation. We propose three specific aims that will use a matrix of charged particles of high energy and atomic number to test important aspects of our hypothesis, the link to cataractogenesis, and		

Task Description:	that will also allow the development of biological countermeasures. Quantitative measurements of the inappropriate commencement of fiber cell differentiation will be made as a function of particle fluence, atomic number and energy, and correlated to the kinetics of early signaling expression in HLE cells in vitro, and in the nascent radiation-induced cataract in the rat in vivo. The human lens is one of the most radiosensitive organs of the body, and is one of the tissues upon which shielding requirements for space flight have been based to avoid cataracts, a late-appearing radiation effect. Recent evidence however indicates that exposure to relatively low doses of space radiation is associated with an increased incidence and earlier appearance of human cataracts than was anticipated. We developed an in vitro model of differentiating human lens epithelial (HLE) cells, and have been studying their response to different types of ionizing radiations prevalent in space travel under conditions that mimicked the normal growth environment in the tissue, but were more amenable to investigation than in vivo models. The objective of our project is to provide evidence showing that particle radiation induces alterations in gene expression profiles when compared to the normal lens differentiation process resulting in premature differentiation subsequently leading to cataract formation. Historically, radiation-induced cataract has been anatomically characterized as posterior sub-capsular (PSC) due to the location of its appearance within the lens after radiation exposure. The mechanisms underlying radiation-induced cataract are thought to involve direct DNA damage to the epithelial cells, which is later expressed as an opacification when the epithelial cells have differentiated into fiber cells and migrated to the PSC location. However lens fiber cells at the juncture of the posterior capsule and suture are terminally differentiated in the region where fibers cells are devoid of nuclear organelles including the cell nucleus. The radiation-induced cataract is a progressive lesion that continues to evolve during the life of the organism, and the molecular basis for the origin and development of radiation-induced cataract are largely unknown. The continuous lifetime migration of germinative epithelial cells down the anterior surface of the lens of the eye, and the progressive transition into primary and secondary fiber cells involves a coordinated symphony of changes. The results from our work will provide information on the important roles of several radiation-responsive signal transduction pathways in lens differentiation and of pre-cataractous changes. Understanding of the molecular basis for radiation-induced cataract pathology will provide important tools to develop biological approaches for countermeasures.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our cell culture model is an exciting tool that demonstrates human lens epithelial cell growth on an extracellular matrix derived from bovine corneal endothelial cells, and also exhibits aspects of fiber cell differentiation with the expression of several morphological and protein markers. The results of our work will be used to pursue countermeasures that could reduce the risk of cataract. These studies may also contribute to estimations of risk assessment for radiation-induced cataracts among astronauts exposed to particles during space travel, especially during missions into deep space. We may also contribute to the development of countermeasures for radiation-induced cataracts from other occupational and clinical radiation exposures. The biological mechanisms underlying normal lens fiber cell differentiation may also be elucidated. The exact factor(s) responsible for the continued replication of lens epithelial cells in embryos and adults remains to be determined. These studies therefore should shed new light on the process of human lens cell differentiation, that has been difficult to explore with animal models in vivo or with human lens cultures in vitro that are either immortalized and/or undergo only limited differentiation.
Task Progress:	We have just begun the final year of our current NASA funding, and have reported here our significant progress on the previous aims to support this new application. In addition to the new information obtained about radiation effects on the differentiating human lens, we have contributed to the basic understanding of normal human lens development and function. The long-term objectives of this 4 year grant is to elucidate the underlying molecular basis for heavy-ion-induced radiation cataractogenesis, and to obtain a more complete understanding of the normal process of lens cell differentiation. The weight of experimental evidence indicates that the germinative epithelium is the primary target for radiation damage leading to cataract. However, there are potentially three hypotheses for the mechanism of radiation-induced cataractogenesis: 1) the increased genotoxic load of radiation damage leads to cataract through a number of intermediate steps involving altered expression, 2) gene expression is altered without genomic changes (the effect here may be at the level of signaling), or 3) the effect is on protein expression directly. There is of course the possibility that these three hypotheses are not mutually exclusive, and that some combination of these hypotheses is involved. The results of our work will be used to pursue countermeasures that could reduce the risk of cataract. These studies may also contribute to estimations of risk assessment for radiation-induced cataracts among astronauts exposed to particles during space travel, especially during missions into deep space. We may also contribute to the development of countermeasures for radiation-induced cataracts from other occupational and clinical radiation exposures. The biological mechanisms underlying normal lens fiber cell differentiation may also be elucidated. The exact factor(s) responsible for the continued replication of lens epithelial cells in embryos and adults remains to be determined. These studies therefore should shed new light on the process of human lens cell differentiation, that has been difficult to explore with animal models in vivo or with human lens cultures in vitro that are either immortalized and/or undergo only limited differentiation. Two articles are in preparation, one to be submitted to Radiation and Environmental Biophysics based on the work presented at the NASA workshop in Moscow and another will be submitted to the Advances in Space Research based on the work to be presented.
Bibliography Type:	Description: (Last Updated: 02/12/2026)
Abstracts for Journals and Proceedings	Blakely EA, Chang PY, Bjornstad KA, Rosen CJ. "Expression of ECM Genes During Human Lens Fiber Cell Differentiation In Vitro." Association of Research in Vision and Ophthalmology (ARVO), Ft Lauderdale, FL, May 2006. Association of Research in Vision and Ophthalmology (ARVO), May 2006. , May-2006

Abstracts for Journals and Proceedings	Blakely EA, Chang PY, Bjornstad KA, Rosen CJ, Bakke J, Puey A, Moncaster J, Mocherla BR, Goldstein LE. "Potential molecular links between DNA damage and ECM remodeling in HZE-irradiated lenses." 4th International Workshop on Space Radiation Research Workshop, Moscow/St. Petersburg, Russia, June 5- 9, 2006. 4th International Workshop on Space Radiation Research Workshop, June 5- 9, 2006. , Jun-2006
Significant Media Coverage	Wysong P. "Astronauts at risk for cortical and posterior subcapsular cataracts. Article about PI Blakely's and L. Chylack's research." Eurotimes (monthly news magazine of the European Society for Cataracts and Refractive Surgery (ESCRS)), August 2006. http://www.escrs.org/PUBLICATIONS/EUROTIMES/06Aug/pdf/Astronautsatrisk.pdf (Accessed 2/10/2017), Aug-2006